

Eshan

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 github.com/eshan-159  leetcode.com/u/eshan159/ 📍 Bhubaneswar

SKILLS

- **Languages:** C++, Python, Go, JavaScript
- **Systems & Concurrency:** Distributed Systems, Concurrency, Multi-threading, Synchronization, Parallel Computing, CUDA, GPU Kernels, Memory Hierarchy, Performance Optimization
- **Cloud & Infra:** AWS EKS, AWS Bedrock, Kubernetes, Terraform, ArgoCD, Apache Spark, VPC, Security Groups, Linux
- **ML & AI:** Deep Learning, Transformers, NLP, VAE, Quantization, Model Compression

EXPERIENCE

- AI Developer Intern, TEXMiN(IIT ISM)**  15 Jun 2025 – 15 Jul 2025
AvatarGPT – Intelligent 3D AI Avatar System
Tech Stack: Python, Unity, PyTorch, Node.js, Rhubarb, Whisper, Sarvam AI
- Deployed production AI inference pipeline on NVIDIA H100 to 70+ enterprise centers — **sub-50ms P99 latency at scale**, containerized via Kubernetes with HPA and load-balanced ingress
 - Deployed to production at **Pitman training** across 70+ centers, enabling users to interact with a live **3D AI avatar** — reducing human support overhead and extending accessible AI-assisted communication to real end-users
- AI Developer Intern, VyomChara**  16 Jul 2025 – 31 Oct 2025
ASL 3D Avatar Interaction Layer:
Tech Stack: Unreal Engine, JavaScript, Node.js
- Built a **real-time text-to-ASL system** (~0.7s latency) with modular Node.js middleware for sign animation sequencing in Unreal Engine.
 - Delivered communication access for **deaf and speech-impaired users**, expanding the product's accessibility reach to a previously underserved population.

PROJECTS

- Phoenix– Self Healing Kubernetes Platform** | https://github.com/eshan-159/Phoenix_Self_Healing_Kubernetes 
Tech Stack: Go, Kubernetes, Terraform, Prometheus, OpenTelemetry, ArgoCD, AWS EKS, Apache Spark, AWS Bedrock
- Built custom Go operators on EKS coordinating distributed control loops with concurrent reconcilers and sliding-window anomaly detection — autonomously recovers from CrashLoops (<5s), memory leaks, and latency spikes (42s under 6× surge) via rollback, pod restart, or predictive scale-up; manages distributed Spark workloads as native K8s resources via Spark Operator
 - Provisioned GitOps infrastructure (Terraform + ArgoCD) with VPC networking, Security Groups, and **K8s network policies**; full observability via Prometheus + OTEL + Loki; chaos-validated with Litmus; AWS Bedrock integration for AI-powered root cause analysis on anomaly events
- Secure OAuth Identity & SSO Server** | <https://github.com/eshan-159/OAuth-social-login> 
• Architected a production OAuth 2.0 + SSO server (Google & GitHub) on normalized PostgreSQL; sustains ~1K–2.5K reads/sec, ~10–15ms latency, and ~500 writes/sec under concurrent multi-threaded load
- Implemented stateless **JWT auth** at ~3K–5K req/sec (<5ms latency); optimized OAuth callbacks to 150–300ms via async processing, rate limiting, and structured logging.
- NovaGEMM – CUDA Matrix Multiplication Engine** | <https://github.com/eshan-159/cuda-gemm-optimization> 
Tech Stack: CUDA, C++, GPU Computing
- Optimized CUDA GEMM engine to near-cuBLAS performance (1465+ GFLOPS), achieving 180× CPU speedup

PROBLEM SOLVING

Codeforces: Expert (Rating 1692) 

LeetCode: 450+ DSA problems on LeetCode(250+ medium, 70+ hard) 

EDUCATION

Institute of Technical Education and Research, Siksha 'O' Anusandhan,
B.Tech in Computer Science Engineering
CGPA: 8.0/10

2023 – 2027
Bhubaneswar, India